

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec:Sr.Super60_NUCLEUS BT **Paper -2(Adv-2022-P2-Model)** **Date: 09-11-2025**

Time: 02.00Pm to 05.00Pm **GTA-05** **Max. Marks: 180**

09-11-2025_Sr.Super60_NUCLEUS-BT_Jee-Adv(2022-P2)_GTA-05_Syllabus,

MATHEMATICS : Total Syllabus
PHYSICS : Total Syllabus
CHEMISTRY : Total Syllabus

Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2022-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type @	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 9 – 14)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 15 – 18)	Questions with Single Correct Choice	+3	-1	4	12
Total				**18**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 27 – 32)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Total				**18**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 45 – 50)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 51 – 54)	Questions with Single Correct Choice	+3	-1	4	12
Total				**18**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

1. 5 different marbles are placed in 5 different boxes randomly. Given that each box can hold any number of marbles. If p is the probability that exactly two boxes remain empty, then the value of $\left[\frac{1}{p}\right]$ is , (where $[.]$ represents GIF).

2. If $\cot^{-1}\left(2^2+\frac{1}{2}\right)+\cot^{-1}\left(2^3+\frac{1}{2^2}\right)+\cot^{-1}\left(2^4+\frac{1}{2^3}\right)+....+\infty=\cot^{-1}x$ then the value of x is _____

3. Let $\bar{a}$ and $\bar{b}$ be two non-collinear vectors such $|\bar{a}|=|\bar{b}|=2 \text{ and } |\bar{a}\times\bar{b}|=|\bar{a}-2\bar{b}|$. If $\bar{r}$ is any vector such that $|\bar{r}|=\frac{4}{\sqrt{3}}$ and $\bar{r}\times(\bar{a}\times\bar{b})=3\bar{a}-2\bar{b}$ then the value of $\left|[\bar{r}\,\bar{a}\,\bar{b}]\right|$ is

4. Suppose $a_1,a_2,a_3,.....a_{100}$ are positive real numbers such that $a_k=\frac{ka_{k-1}}{a_{k-1}-(k-1)}$ For k = 2, 3 .....100 Given that $a_{20}=a_{23}$ then sum of the digits of a_{100} is equal to

5. A certain function $f:[1,\infty)\to(-\infty,\infty)$ has the properties that $f(3x)=3f(x)$ for all $x\geq 1$ and that $f(x)=1-|x-2| \text{ for } 1\leq x\leq 3$, N be the smallest positive number that satisfies $f(N)=f(2001)$, then $\left[\frac{N}{30}\right]$ is equal to (where $[x]$ denotes greatest integer less than or equal to)

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6. Let f (x) be a non-constant thrice differentiable function defined on $(-\infty,\infty)$ such that $f(x)=f(6-x)$ *and* $f'(0)=0=f'(2)=f'(5)$ If n is the minimum number of roots of $\left(f''(x)\right)^2+f'(x)f'''(x)=0$ in the interval[0,6], then the value of n/2 is

7. Let f : $R\to R$ be a function defined as f (x+y) = f(x) + f(y) - 3xy(x+y) - 1 $\forall x,y\in R$, $f'(0)=-1$. Find the least positive integer value of x which satisfies the inequality $1-f(x)-f^3(x)>f(1-5x)$.

8. Let ω be a non real complex number satisfying $\omega^{23}=1$. If $N=\sum_{k=1}^{22}\frac{1}{1+\omega^{8k}+\omega^{16k}}$ then sum of digits in N is equal to

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

9. If f(x) = 2 + |x - 1| & g(x) = min {f(t) / $x\le t\le x^2+x+1$}, then g(x) is

A) Continuous on R **B)** Not differentiable at x = 0, 1

C) Not differentiable at x = -1 **D)** Not continuous at x = 1

10. Normals at P and Q to the parabola $y^2=8x$ intersect at R(8,−12) and tangents at these points P & Q are intersect at T. Then select the correct alternative(s)

A) area of triangle TPQ is 2

B) slope of chord PQ is $\frac{2}{3}$

C) If 'θ' is angle between normal at P and Q, then tan θ is 3

D) Slope of chord PQ is $\frac{3}{2}$

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11. Which of the following is true about the equation:

$$\sqrt{x^3+3x^2+3x+9}=2+\sqrt[3]{x^2+6x+1}$$

A) Sum of roots is -2

B) Sum of product of roots taken two at a time is -3

C) Sum of magnitude of roots is 3

D) Sum of magnitude of roots is 4

12. $f:[a,b]\to[5,6]$ be defined as $y=f(x)=2x^3-9x^2+12x+1$. Then which of the following is true?

A) Minimum value of a for y=f(x) to be a one-one function is $\frac{1}{2}$

B) Maximum value of a for y=f(x) to be a one-one function is 2

C) Minimum value of b for y=f(x) to be an onto function is 1

D) Maximum value of b for y=f(x) to be an onto function is $\frac{5}{2}$

13. If $\int_0^{\frac{\pi}{2}} e^x(\sin x)dx=A_1$, Then which of the following is/are true?

A) $A_1\le 2\sqrt{e^\pi}$

B) $A_1\ge\frac{\left(\sqrt{e^\pi}\right)\pi}{2}$

C) $A_1\le\left(\sqrt[4]{\frac{e^\pi}{4}}+\sqrt{e^\pi}\right)\frac{\pi}{4}$

D) $A_1\le\sqrt{e^\pi}\,\frac{\pi}{4}$

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14. Which of the following is/are INCORRECT

A) The constant term in the expansion of $\left(2x+\frac{1}{x^7}+3x^2\right)^5$ is 1080

B) A group of 40 students appeared in an examination of 3 subjects - mathematics, physics and chemistry. It was found that all students passed in at least one of the subjects. 20 students passed in mathematics, 25 students passed in physics, 16 students passed in chemistry, at most 11 students passed in both mathematics and physics, at most 15 students passed in both physics and chemistry, at most 15 students passed in both mathematics and chemistry. Then maximum number of students passed in all the three subjects is 10

C) If $T_n=\frac{(2n+1)\left(2n^2+2n-1\right)}{(n-1)^2n^2(n+1)^2(n+2)^2}, n\in N, n\geq 2$. If $k=\lim\limits_{n\to\infty}\sum\limits_{r=2}^{n}T_r$ Then k is 3

D) Let $a_1,a_2.....a_{10}$ be 10 observations such that $\sum_{k=1}^{10}a_k=50$ and $\sum_{1\leq k<j\leq 10}a_k.a_j=1100$. Then the standard deviation of $a_1,a_2.....a_{10}$ is equal to 5

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

15. If $\sum_{k=1}^{n}\frac{k^2-\frac{1}{2}}{k^4+\frac{1}{4}}=\frac{an^2}{bn^2+cn+1}$ Then

A) $a+b+c=6$ **B)** $a+2b+3c=6$

C) $3a+2b+c=6$ **D)** $a^2+b^2+c^2=6$

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16. There are two dice A and B both having six faces. Die A has three faces marked with 1, two faces with 2 and one face marked with 3. Die B has one face marked with 1, two faces marked with 2 and the three faces marked with 3. Both dies are thrown randomly once. Let E be the event of getting sum of the numbers appearing on top faces equal to x, P (E) is maximum if x is...

A) 3 **B)** 4 **C)** 5 **D)** 6

17. If $\overline{a_1} = a\hat{i} + b\hat{j}, \overline{a_2} = \overline{a_1} - \left(2\hat{i} + \sqrt{5}\hat{j}\right)$ and $\overline{a_3} = \overline{a_1} + \left(2\hat{i} - \sqrt{5}\hat{j}\right)$ then

$$\begin{vmatrix} 1+\overline{a_1}.\overline{a_1} & 1+\overline{a_1}.\overline{a_2} & 1+\overline{a_1}.\overline{a_3} \\ 1+\overline{a_2}.\overline{a_1} & 1+\overline{a_2}.\overline{a_2} & 1+\overline{a_2}.\overline{a_3} \\ 1+\overline{a_3}.\overline{a_1} & 1+\overline{a_3}.\overline{a_2} & 1+\overline{a_3}.\overline{a_3} \end{vmatrix}$$ is equal to

A) 0 **B)** 20 **C)** 40 **D)** 80

18. The value of the definite integral $\int_0^{\frac{\pi}{2}} \frac{\sin 5x}{\sin x} dx$ is :

A) 0 **B)** $\frac{\pi}{2}$ **C)** π **D)** 2π

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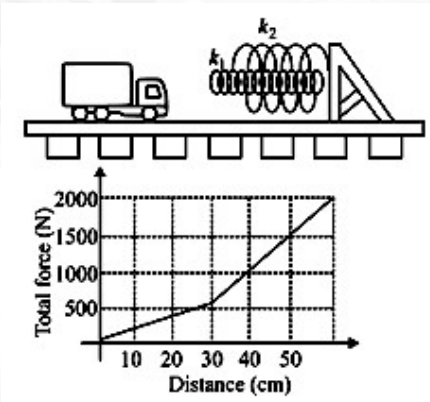
PHYSICS **Max Marks: 60**

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
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19. A 2144 kg freight car roles along rails with negligible friction. The car is brought to rest by a combination of two coiled springs as illustrated in Figure. Both springs are described by Hooke's law with $k_1 = 1600\,Nm^{-1}$ and $k_2 = 3400\,Nm^{-1}$. After the first spring compresses a distance of 30.0 cm, the second spring acts with the first to increase the force as additional compression occurs as shown in the graph . The car comes to rest 50.0 cm after first contracting the two-spring system. Find the car's initial speed (in $\times 10^{-1} ms^{-1}$).

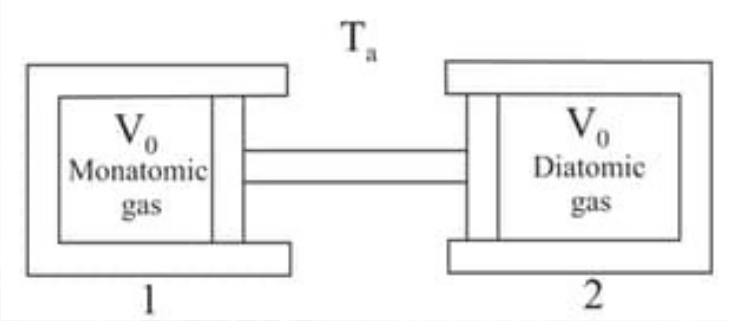


20. The radioactivity of a sample is R_1 at a time T_1 and R_2 at a time T_2. If the half-life of the specimen is T, the number of atoms that have disintegrated in the time $(T_2 - T_1)$ is equal to $\frac{n(R_1 - R_2)T}{ln4}$. Here n is some integral number.

What is the value of n?

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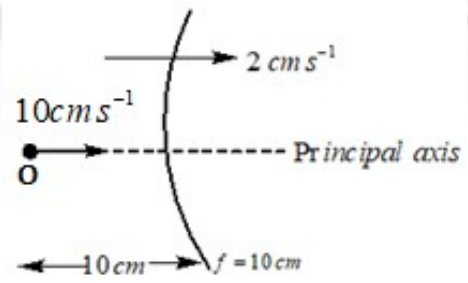
21. The two conducting cylinder-piston systems shows below are linked. Cylinder 1 is filled with a certain molar quantity of a monatomic ideal gas, and cylinder 2 is filled with an equal molar quantity of a diatomic ideal gas. The entire apparatus is situated inside an oven whose temperature is $T_a = 27^o C$. The cylinder volumes have the same initial value V_0 = 100 cc. When the oven temperature is slowly raised to $T_b = 127^o C$. What is the volume change ΔV (in cc) of cylinder 1?



22. The energy of a system as a function of time t is given as E(t) = $A^2\ e^{-\alpha t}$, where $\alpha = 0.2 s^{-1}$. The measurement of A has an error of 1.25%. If the error in the measurement of time is 1.50%, the percentage error in the value of E(t) at t=5 s is

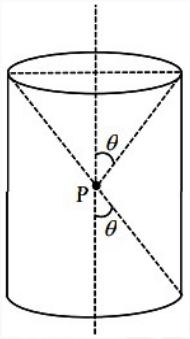
23. An ac ammeter is used to measure current in a circuit. When a given direct current passes through the circuit, the ac ammeter reads 3 A. When another alternating current passes through the circuit, the ac ammeter reads 4 A. Then find the reading of this ammeter (in A), if dc and ac flow through the circuit simultaneously.

24. What is the velocity (in $cm\,s^{-1}$) of image (with respect to ground) in situation shown in Figure. (O =object, f= focal length). Object moves with velocity 10 cm/s and mirror moves with velocity 2 $cm\,s^{-1}$ as shown.



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25. A charge is kept at the central point P of a cylindrical region. The two edges subtend a half-angle θ at P, as shown in the figure. When θ = 30°, then the electric flux through the curved surface of the cylinder is Φ If θ=60°, then the electric flux through the curved surface becomes $\Phi/\sqrt{n}$, where the value of n is_____



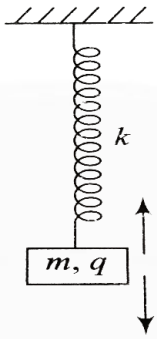
26. A highly rigid cubical block B_1 of small mass m and side L is fixed rigidly onto another cubical block B_2 of same dimensions and of low modulus of rigidity η, such that the lower face of B_1 completely covers the upper face of B_2. The lower face of B_2 is rigidly held on a horizontal surface. A small force F is applied perpendicular to one of the side faces of B_1. After force is withdrawn, block B_1 executes small oscillations, the time period of which is $\pi\sqrt{\frac{\alpha m}{\eta L}}$ The value of α is_____

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
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Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

27. The following figure shows a block of mass m suspended from a fixed point by means of a vertical spring. The block is oscillating simple harmonically and carries a charge q. There also exists a uniform electric field in the region. Consider four different cases. The electric field is zero in case 1, mg/q downward in case 2, mg/q upward in case 3, and 2mg/q downward in case 4. The speed at mean position of block is same in all cases. Select the correct alternative(s).

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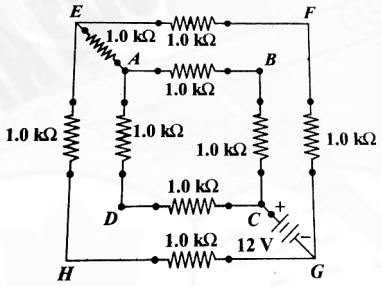
A) Time periods of oscillation are equal in case 1 and case 3.

B) Amplitudes of displacement are same in case 2 and case 3.

C) The maximum elongation (increment in length from natural length) is maximum in case 4.

D) Time periods of oscillation are equal in case 2 and case 4.

28. In the given circuit



A) The equivalent resistance between C and G is $3k\Omega$

B) The current provide by the source is 4 m A

C) The current provide by the source is 8 mA

D) Voltage across point E and G is 4V

29. A small conducting sphere of radius a mounted on an insulated handle and given a positive charge q inserted through a very small hole in the wall of a hollow conducting sphere of inner radius b and outer radius c. The hollow sphere is supported on an insulating stand and is initially uncharged. The small sphere is placed at the center of the hollow sphere. Neglect any effect of the hole. Which of the following statements will be true for this system?

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A) No work will be done in carrying a small charge from the inner conductor to the outer conductor.

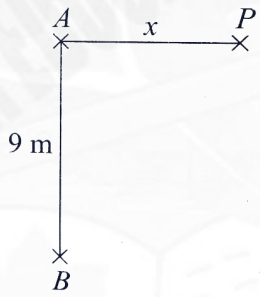
B) The electric field at a point in the region between the spheres at a distance r from the centre is equal to $q / 4\pi\varepsilon_0 r^2$.

C) The electric field at a point outside the hollow sphere at a distance r from the center is $q / 4\pi\varepsilon_0 r^2$

D) The potential of the inner sphere with respect to the outer sphere is given by

$$V_{ab} = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)$$

30. A radio transmitting station operating at a frequency of 120 MHz has two identical antennas that radiate in phase. Antenna B is 9 m to the right of antenna A. Consider point P at a horizontal distance x to the right of antenna A as shown in Figure The value of x and order for which the constructive interference will occur at point P are



A) x = 14.95 m. n=1 **B)** x= 5.6m, n=2

C) x= 1.65 m, n=3 **D)** x= 0 m, n=3.6

31. A uniform capillary tube of inner radius r is dipped vertically into a beaker filled with water. The water rises to a height h in the capillary tube above the water surface in the beaker. The surface tension of water is σ. The angle of contact between water and the wall of the capillary tube is θ. Ignore the mass of water in the meniscus. Which of the following statements is (are) true?

A) For a given material of the capillary tube, h decreases with increase in r

B) For a given material of the capillary tube, h is dependent of σ

C) If this experiment is performed in a lift going up with a constant acceleration, then h decreases

D) h is proportional to contact angle θ.

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32. A bubble has surface tension S. The ideal gas inside the bubble has ratio of specific heats $\gamma = 5/3$. The bubble is exposed to the atmosphere and it always retains its spherical shape. When the atmospheric pressure is P_{a_1}, the radius of the bubble is found to be r_1, and the temperature of the enclosed gas is T_1. When the atmospheric pressure is P_{a_2}, the radius of the bubble and the temperature of the enclosed gas are r_2 and T_2, respectively. Which of the following statement(s) is (are) correct?

A) If the surface of the bubble is a perfect heat insulator, then $\left(\frac{r_1}{r_2}\right)^5 = \frac{P_{a_2} + \frac{2s}{r_2}}{P_{a_1} + \frac{2s}{r_1}}$

B) If the surface of the bubble is a perfect heat insulator, then the total internal energy of the bubble including its surface energy does not change with the external atmospheric pressure.

C) If the surface of the bubble is a perfect heat conductor and the change in atmospheric temperature is negligible, then $\left(\frac{r_1}{r_2}\right)^3 = \frac{P_{a2} + \frac{4s}{r_2}}{P_{a1} + \frac{4s}{r_1}}$

D) If the surface of the bubble is a perfect heat insulator, then $\left(\frac{T_2}{T_1}\right)^{\frac{5}{2}} = \frac{P_{a_2} + \frac{4s}{r_2}}{P_{a_1} + \frac{4s}{r_1}}$

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

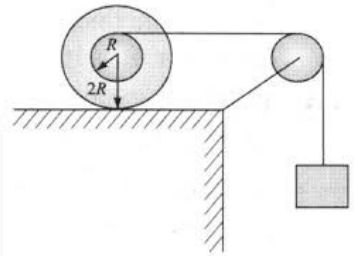
This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

33. In the figure mass of both spherical body and block is m. Moment of inertia of the spherical body about center of mass is $2mR^2$. The spherical body rolls on the horizontal surface. There is no slipping at any surfaces in contact. The ratio of kinetic energy of the spherical body to that of block is

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A) $\frac{3}{4}$ B) $\frac{1}{3}$ C) $\frac{2}{3}$ D) $\frac{1}{2}$

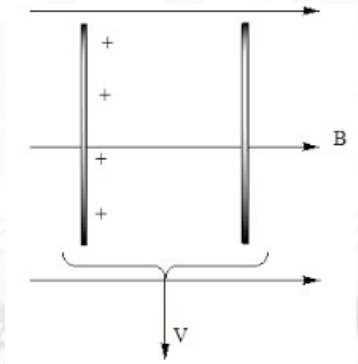
34. A plane light wave of intensity I=0.20 W cm^{-2} fall on a plane mirror surface with reflection coefficient $\rho = 0.8.$ The angle of incident is 45°. In terms of corpuscular theory, find the magnitude of the normal pressure exerted on the surface.

A) $1.2nN\,cm^{-2}$ B) $0.2nN\,cm^{-2}$ C) $2.6nN\,cm^{-2}$ D) $0.6nN\,cm^{-2}$

35. Which of the following product of e, h, μ, G (where μ is the permeability) be taken so that the dimensions of the product are same as that of the speed of light?

A) $he^{-2}\mu^{-1}G^{0}$ B) $h^{2}eG^{0}\mu$ C) $h^{0}e^{2}G^{-1}\mu$ D) $hGe^{-2}\mu^{0}$

36. A parallel plate capacitor is moving with a velocity of $25\,ms^{-1}$ through a uniform magnetic field of 4.0 T as shown in figure. If the electric field within the capacitor plate is $400\,NC^{-1}$ and plate area $25\times10^{-7}\,m^{2}$, then the magnetic force experienced by the positive charge plate is



A) $8.85\times10^{-13}\,N$ directed out of the plane of the paper

B) Zero

C) $8.85\times10^{-15}\,N$ directed out of the plane of the paper

D) None of the above

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CHEMISTRY **Max Marks: 60**

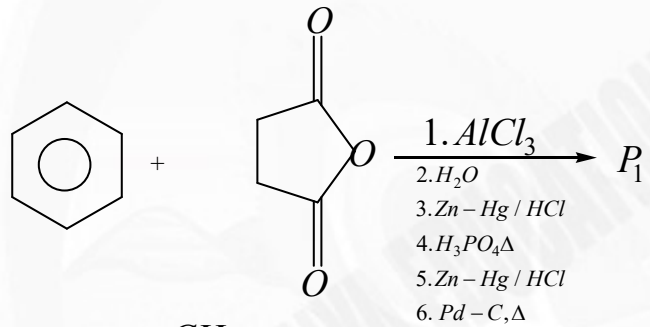
SECTION-I
(INTEGER ANSWER TYPE)

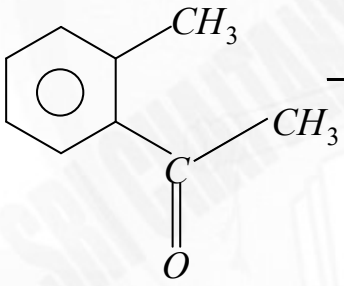
- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

37. In how many of the following species bond order is not an integer BF_3, O_3, graphite, diamond, carborundum(SiC), NO_3^-, CO_3^{2-}, SO_3^{2-}, ClO_3^- _____

38. Consider the following synthesis of a fused polynuclear aromatic hydrocarbon





$$P_2 \xrightarrow[2.H_3O^+]{1.Alk.KMnO_4} P_3 \xrightarrow{\Delta} P_4$$

$$P_1 + P_4 \xrightarrow[\substack{2.Polyphosphoric\ acid \\ 3.Zn-Hg/HCl \\ 4.Pd-C,\Delta}]{1.AlCl_3} P_5$$

Report the value of X+Y-Z = _____
Where
[X- Degree of unsaturation in P_5
Y-Number of tertiary carbons in P_5
Z-Number of Hydrogens in P_5]

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39. Consider following statements/reactions:

(i) $F_2 + H_2O \rightarrow$ fluorine containing compound+ gas A+ gas B, only gas A is paramagnetic

(ii) NaI + conc. $H_2SO_4 \rightarrow$X

(iii) KI enchances solubility of X in water.

Now consider the following reaction:

Gas B + dry X → product

How many moles of oxygen atoms are present in one mole of product?

40. The conductance of a 0.0015 M aqueous solution of a weak monobasic acid was determined by using a conductivity cell consisting of platinized Pt electrodes. The distance between the electrodes is 120 cm with an area of cross section of 1 cm^2. The conductance of this solution was found to be 5×10^{-7} S. The pH of the solution is 4.The value of limiting molar conductivity (Λ_m^0) of this weak monobasic acid in aqueous solution is $Z \times 10^2$ S cm^2 mol^{-1}. The value of Z is

41. Consider the following reaction $2H_{2(g)} + 2NO_{(g)} \rightarrow N_{2(g)} + 2H_2O_{(g)}$

Which follows the mechanism given below

$$2NO_{(g)} \underset{K_{-1}}{\overset{K_1}{\rightleftharpoons}} N_2O_{2(g)} \quad \text{(Fast equilibrium)}$$

$$N_2O_{2(g)} + H_{2(g)} \xrightarrow{K_2} N_2O_{(g)} + H_2O_{(g)} \quad \text{(Slow reaction)}$$

$$N_2O_{(g)} + H_{2(g)} \xrightarrow{K_3} N_{2(g)} + H_2O_{(g)} \quad \text{(Fast reaction)}$$

The order with respect to H_2 is

42. $$Na_2Cr_2O_{7(s)} \xrightarrow[H_2SO_4,\Delta]{NaCl_{(s)}} coloured\ product(p)$$

$$n-hep\tan e \xrightarrow[at\ High\ temperature\ \&\ pressure]{Cr_2O_3} (Q)$$

$$(P)+(Q) \xrightarrow[ii)H_3O^+]{i)CS_2} (R)$$

The total number of sp^2 hybridized carbon atoms present in one molecule of compound (R) is

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43. Compound (A) with Molecular weight 108 contained 88.89% C and 11.11% H. It gave a white precipitate with ammonical silver nitrate solution. Complete hydrogenation of (A) gave another compound (B) with molecular weight 112. Oxidation of (A) gave an acid (C) and other side product. Decarboxylation of this acid gave cyclohexane.

$$(B) \xrightarrow[(monobromination)]{Br_2/h\upsilon(sun\,light)} \xrightarrow[\Delta]{t-BuO^-K^+} \text{Major product alkene (D)}$$

Number of α -H (alpha hydrogens) present in one molecule of compound (D) is:

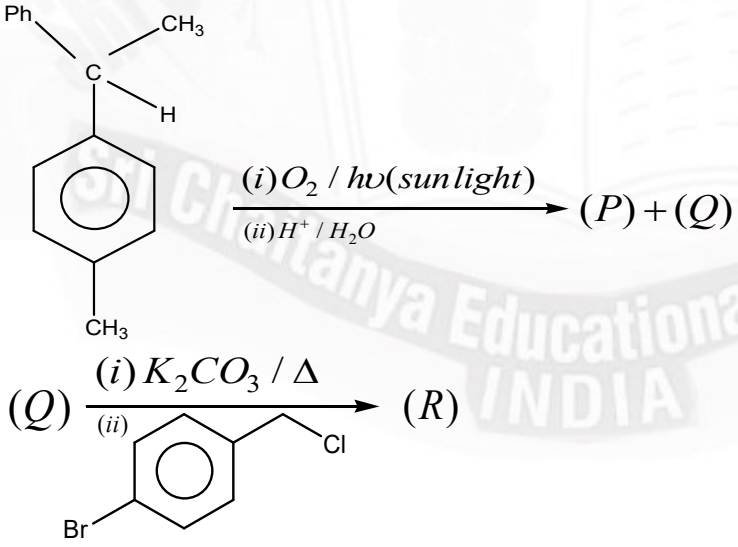
44. An organic compound P having molecular formula $C_6H_6O_3$ gives ferric chloride test and does not have intramolecular hydrogen bond. The compound P reacts with 3 equivalents of NH_2OH to produce oxime Q Treatment of P with excess methyl iodide in the presence of KOH produces compound R as the major product.

The total number of methyl $(-CH_3)$ group(s) in compound R is

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

45. The correct statement(s) about the following reaction sequence is are



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Given: Degree of Unsaturation of (P) is greater than (Q)

A) (P) gives yellow precipitate with $Br_2 / NaOH$

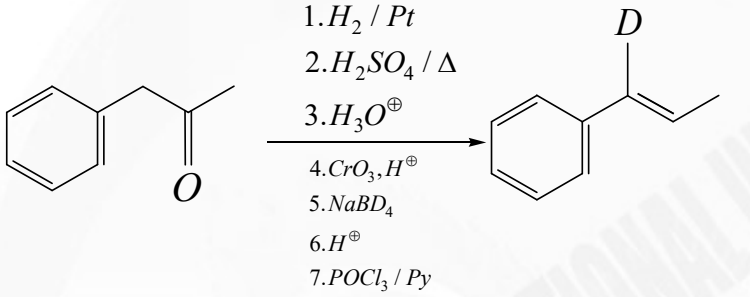
B) (Q) and (R) can be differentiated by neutral $FeC\ell_3$ aqueous solution

C) (R) gives precipitate with $AgNO_{3(aq)}$

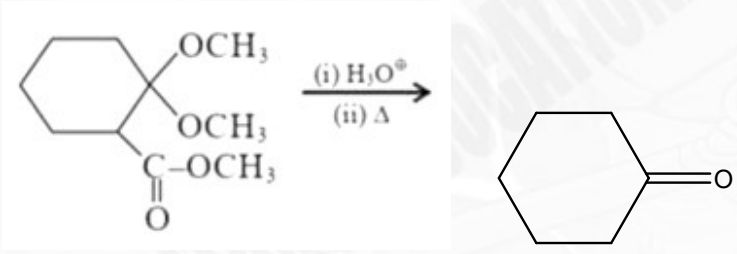
D) (P) gives coloured precipitated with Brady's reagent

46. Identify the option in which product is correct method

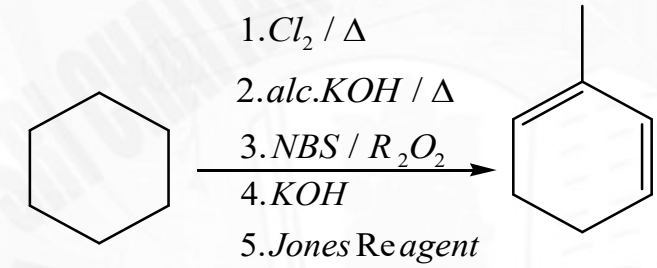
A)



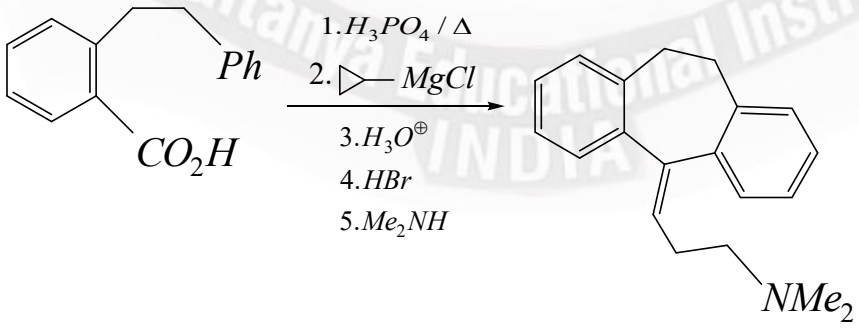
B)



C)



D)



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47. Among the following, the correct statement(s) is/are:

A) The two six-membered cyclic hemiacetal forms of D-(+)-glucose are called as C2 epimers.

B) Invert sugar is prepared by acid catalyzed hydrolysis of maltose.

C) DNA contains cytosine and thymine as pyrimidine bases.

D) Glycine is an optically inactive amino acid.

48. Some standard electrode potentials at 298 K are given below :

$Pb^{2+}/Pb = -0.13$ V

$Ni^{2+}/Ni = -0.24$ V

$Cd^{2+}/Cd = -0.40$ V

$Fe^{2+}/Fe = -0.44$ V

To a solution containing 0.001 M of X^{2+} and 0.1 M of Y^{2+}, the metal rods X and Y are inserted (at 298 K) and connected by a conducting wire. This resulted in dissolution of X. The correct combination(s) of X and Y, respectively, is (are):

(Given: Gas constant, R = 8.314 J K^{-1} mol^{-1}, Faraday constant, F = 96500 C mol^{-1})

A) Cd and Ni **B)** Cd and Fe **C)** Ni and Pb **D)** Ni and Fe

49. The correct statement(s) about surface properties is (are):

A) Adsorption is accompanied by decrease in enthalpy and decrease in entropy of the system.

B) The critical temperatures of ethane and nitrogen are 563 K and 126 K, respectively. The adsorption of ethane will be more than that of nitrogen on same amount of activated charcoal at a given temperature.

C) The Tyndall effect is due to the fact that colloidal particles does not scatter light in all directions.

D) Brownian motion of colloidal particles does not depend on the size of the particles but depends on viscosity of the solution.

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50. Mixture(s) or solutions showing positive deviation from Raoult's law at 35°C is (are)

A) carbon tetrachloride + methanol

B) carbon disulphide + acetone

C) benzene + toluene

D) phenol + aniline

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

51. Which of the following combination of solution forms buffer?

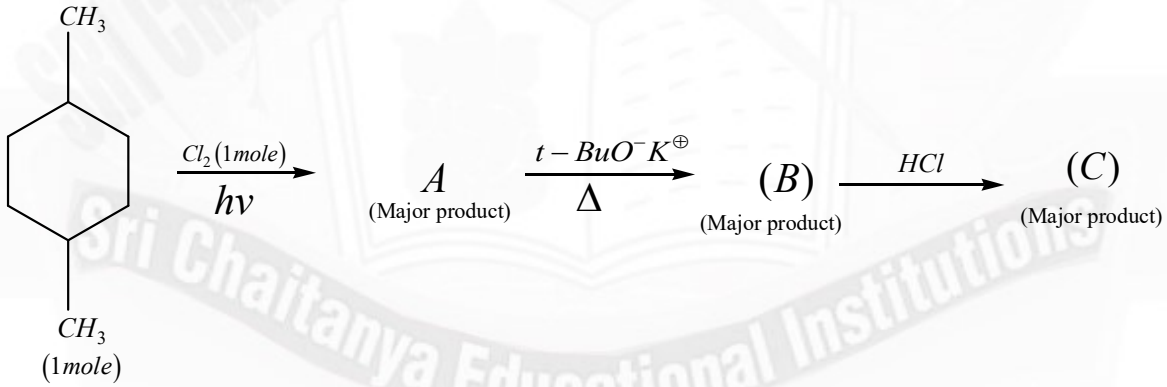
A) HNO_3 *and* CH_3COOH

B) $NaNO_3$ *and* CH_3COONa

C) $HClO_4$ *and* $KClO_4$

D) CH_3COOH *and* CH_3COONa

52. Consider the following reaction sequence the correct statement is:



A) Product A and C is same

B) Number of $\alpha - H$ in major product B is 2

C) product C is optically active

D) major product C has one chiral center

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53. Among the following complexes/molecules the total number of diamagnetic species is/are_____

$\left[Mn(NH_3)_6\right]^{3+}, \left[MnCl_6\right]^{3-}, \left[Ni(CO)_4\right], \left[CoF_6\right]^{3-}, \left[Fe(NH_3)_6\right]^{3+},$

$O_3, BaO_2, \left[Pt(NH_3)_2 Br_2\right], \left[Co(en)_3\right]Cl_2$ and $\left[Fe(CN)_6\right]^{4-}$

A) 5 **B)** 4 **C)** 6 **D)** 7

54. Which one of the following statement is INCORRECT about O_3

A) O_3 oxidises black PbS into white $PbSO_4$

B) It tarnishes silver metal

C) It is a meta stable allotropic form of oxygen

D) H_2O_2 can oxidise O_3

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